Exploring Asymmetry in Walkability Perception: Evidence from 2D Video Experiments in Latin American Cities

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Abstract. This study explores asymmetries in perceived walkability, examining whether improvements and deteriorations in urban attributes exert different effects on pedestrians' perceptions. Using video-based experiments designed under a one-factor-at-a-time approach, participants from three Latin American cities (Porto Alegre (Brazil), Barranquilla (Colombia), and Concepción (Chile)) assessed simulated streetscapes that varied in sidewalk conditions, safety, security, destinations, and attractiveness. Responses from 1,726 participants were analyzed using a pooled ordered logit model with city-specific interactions. Results reveal that deteriorations have substantially larger negative effects than the corresponding positive effects of improvements, indicating a clear pattern of loss aversion consistent with Prospect Theory. Sidewalk conditions, security, and attractiveness emerged as the most influential attributes shaping walkability perceptions, with the strongest penalties for deteriorations observed in Porto Alegre and the highest gains from improvements in Concepción. These findings highlight that enhancing walkability requires not only adding positive features but also preventing the degradation of environmental attributes. The simple OFAT-based video approach proved to be an accessible and cost-effective tool for supporting evidence-based urban design and transport planning in diverse contexts.

Keywords: Walkability, Loss aversion, Ordered logit model, Video-based experiment.

1 Introduction

Walkability refers to the capacity of the urban environment to support and encourage walking, influenced by different physical attributes of the built environment as well as by aspects related to public and traffic safety (Tiang and Ewing, 2017; Larranaga et al., 2018; Arellana et al., 2020). Most studies assume symmetric effects of these attributes, implying that an improvement and a deterioration of equal magnitude produce opposite but equivalent impacts. Even advanced discrete choice models commonly adopt this assumption. However, evidence from behavioral economics, particularly Prospect

Theory (Kahneman and Tversky, 2013), demonstrates that individuals often perceive losses more intensely than gains, a phenomenon known as loss aversion.

In the context of walkability, limited research has examined such asymmetries. Larranaga et al. (2018), for instance, applied the best–worst scaling technique to analyze barriers and incentives to walkability in Porto Alegre (Brazil), identifying significant differences between the evaluation of improvements and deteriorations in built environment attributes. Their results showed that positive features (such as good connectivity, adequate pavement, and public and traffic safety) exerted stronger impacts than their negative counterparts. Similarly, Ruiz-Padillo et al. (2022) identified unequal roles of incentives and barriers to walking in a smaller Brazilian city (Cachoeira do Sul), showing that removing obstacles and adding improvements yield distinct outcomes. Dong et al. (2019), in a study conducted in Harbin (China), estimated machine learning models to assess the relative importance of various walkability attributes. The authors identified that the absence of certain attributes generated dissatisfaction, while their presence provided limited additional satisfaction. These findings suggest that gains and losses related to urban attributes are not perceived equally. The concept of loss aversion reinforces this idea and, moreover, indicates that people tend to penalize losses more intensely.

This study aims to deepen understanding of asymmetric perceptions of walkability by evaluating how improvements and deteriorations in urban attributes are valued in three Latin American cities - Barranquilla (Colombia), Concepción (Chile), and Porto Alegre (Brazil) - using 2D video-based experiments. Through simple experimental design and direct analyses, feasible in any statistical software, the study seeks to provide practical and accessible tools to support public managers in prioritizing urban improvements.

2 Methodology

To assess the impact of urban attributes on walkability perception, an experimental survey was designed using 2D videos for three Latin America cities: (i) Porto Alegre, Brazil (POA), (ii) Barranquilla, Colombia (BAQ) and (iii) Concepción, Chile (CON). The scenarios were also developed to be shown through virtual reality, but those weren't applied to this research. The study followed a One Factor at a Time (OFAT) design in which only one altered attribute (to better or worse), while others remained at their baseline level. This simplified approach allows the effect of each improvement to be estimated objectively, facilitating its application by public managers in the prioritization of interventions.

Data was collected online using videos that simulated a walk from the pedestrian's perspective (example in Figure 1). Volunteers watched the baseline scenario prior to improved and worsened scenarios. Participants were not informed about which attribute had been modified. After viewing each video, participants assigned scores from 1 (very poor) to 10 (excellent), reflecting their assessment of walking in the displayed street.

The attributes used were the same used on Facchini et al. (2025), who modeled fictitious scenarios to evaluate the use of virtual reality in walkability studies using 5

attributes (shown in Table 1). The good and poor levels were defined by improving or worsening each attribute from the baseline (regular) condition. A focus group with 27 participants was conducted in POA to validate the videos and evaluate potential improvements. Results indicated that, in most cases, differences between scenarios were perceived regardless of the display device (TV, monitor, or smartphone) or presentation format (individual or paired videos). However, improvements related to the representation of *Security* were suggested. Since the changes in each level were the same for all cities, the adjustments were applied for all of them.

Table 1. Attributes, factors and levels

Attribute	Factor	Poor	Regular	Good
Sidewalk conditions	Sidewalk width	Narrow (1.5 m)	Regular (3 m)	Wide (5 m)
	Pavement quality	Poor	Medium	Good
Safety	Vehicle speed	60 km/h	50 km/h	25 km/h
	Traffic volume	70%	25%	5%
	Crosswalks	No	Yes	Yes (with raised crosswalk)
	Traffic lights/signs	No	Yes	Yes
Security	Pedestrian flow	0%	10%	20%
	Presence of guards/police	0	0	2 guards and 1 police car
	Public lighting	Lighting poles at the beginning and end of the block	Lighting poles at the beginning, middle, and end of the block	Lighting poles and building lighting
Destinations	Commercial/institutional density	Residential only	Two commercial buildings	8 to 9 commercial and institutional buildings
	Transport stops	No	Yes	Yes
Attractiveness	Presence of trees	No	Some trees	Many trees
	Building and street aesthetics	Poor, deteriorated	Regular	Good
	Cleanliness	Very dirty	Slightly dirty	Clean

The baseline scenario was developed to represent a downtown street in each city, allowing participants to imagine themselves walking in this environment. Visual modeling was carried out using Twinmotion, based on Google Street View images, and included typical urban elements and buildings from each city, such as stores and buses.

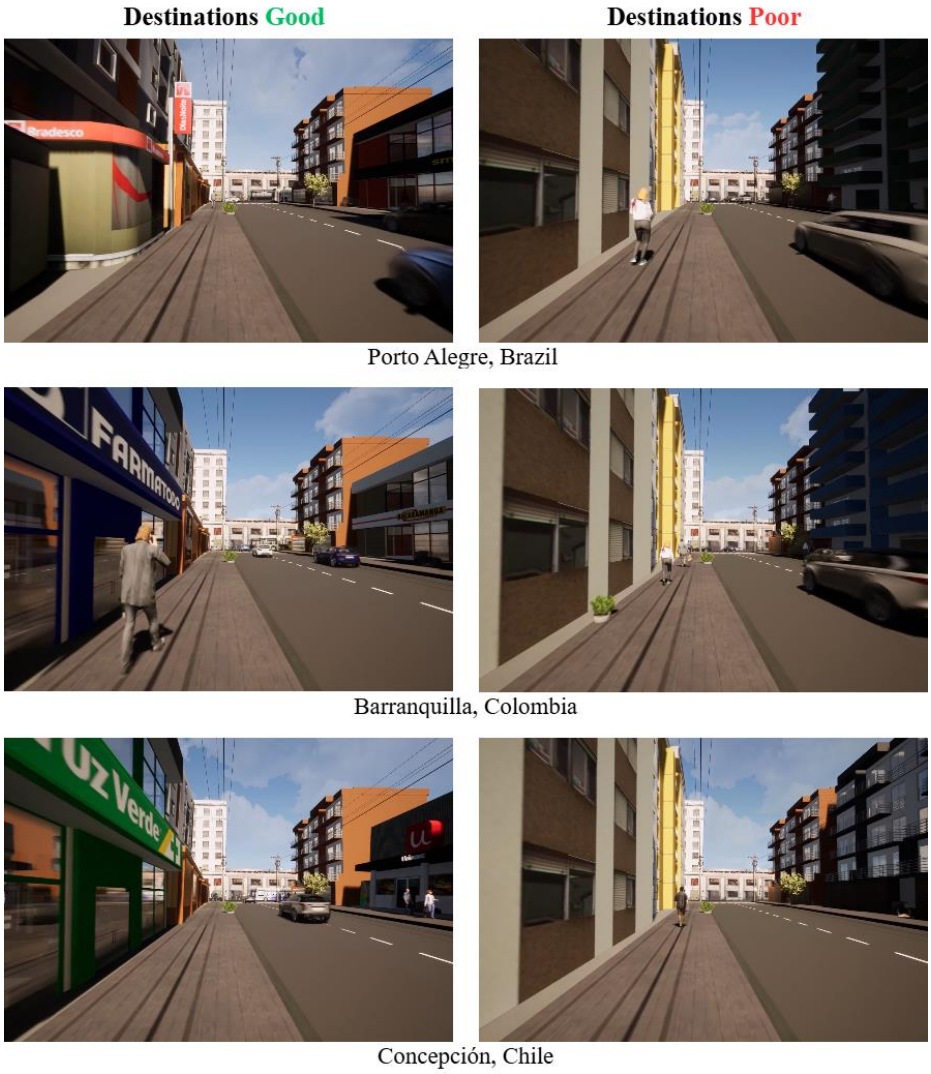


Fig. 1. Example of the presented scenarios (link to access the videos).

To analyze the data, we estimated a pooled cumulative ordered logit model for the three cities. We included one dummy for Barranquilla and one for Concepción (Porto Alegre as the reference city). Improvements and deteriorations were entered into separate effect-coded variables (−1/0/+1), and each attribute (improved and worsened) was interacted with the city dummies to allow city-specific estimates and enable statistical comparison of parameters in a joint model.

The estimation was performed in the R environment using the `polr()` function from the MASS package. The model assumes a continuous latent propensity influenced by the scenario attributes, mapped to the observed 1–10 scores via ordered thresholds

(Hensher and Greene, 2011). In this specification, the effect of attribute X_j in Barranquilla is $\beta_j + \delta_{j,\text{BAQ}}$ and in Concepción is $\beta_j + \delta_{j,\text{CON}}$. Here, β_j refers to main-effect coefficient for attribute X_j in the reference city (Porto Alegre), and $\delta_{j,\text{BAQ}}$ and $\delta_{j,\text{CON}}$ are the city-specific increments arising from the interaction terms $X_j \times \text{BAQ}$ and $X_j \times \text{CON}$, respectively.

We tested equality with Porto Alegre via pairwise t-tests on the city–attribute interactions (H_0: $\delta_{j,\text{city}} = 0$) and compared the two non-reference cities via joint Wald tests of the linear contrast (H_0: $\delta_{j,\text{BAQ}} - \delta_{j,\text{CON}} = 0$). We also applied additional tests (Wald) to assess if interactions that showed no difference on t-tests could be removed from the model.

Based on the predicted category probabilities (scores 1–10) from the ordered logit, we computed the expected score for each scenario as a probability-weighted average of the scale (Eq. 1).

$$\mathbb{E}[Y \mid x] = \sum_{k=1}^{10} k \cdot \Pr(Y = k \mid x) \quad (1)$$

where, Y_{is} the assigned score (1–10), x denotes the scenario attributes/covariates (e.g., sidewalks, safety), $\Pr(Y = k \mid x)$ is the predicted probability of score $k$given those attributes, and $\mathbb{E}[Y \mid x]$is the expected (mean) score.

3 Results

In total, 625 responses were collected in POA; 907 responses in BAQ and 194 responses in CON. Prior to model estimation, we performed response-quality checks and excluded cases showing response patterns inconsistent with the comparative task (e.g., 10 for the baseline and 1 for all modified scenarios, or systematically higher ratings for worsened than improved scenarios). In total, 27 cases in POA and 52 in BAQ were excluded.

Table 2 presents the results of the pooled ordered logit model estimated. In general, the coefficients for the improvement and worsening of each attribute show the expected direction: improvements have positive effects on the perceived evaluation, whereas deteriorations have negative impacts of greater magnitude. Some attributes were not statistically significant at the 95% confidence level ($p < 0.05$), all of them for improved scenarios: *Sidewalk conditions* for BAQ and CON, *Safety* and *Destinations* for POA and *Security* for CON. These results suggest that in some cases, the perceived changes were not strong or consistent enough among participants to generate statistically significant variations.

The coefficient associated with the dummy for Barranquilla indicates that, when all attributes are held constant (baseline scenario), the average score given by respondents from Barranquilla tends to be slightly lower than that of Porto Alegre. Similarly, the dummy for Concepción suggests that respondents from Concepción also tend to assign lower scores than those from Porto Alegre, and the difference is somewhat larger. Comparing Barranquilla and Concepción directly, tests on the BAQ–CON contrast found differences only for *Safety* (improved), *Destinations* and *Attractiveness* (both improved

and worsened); the remaining attributes showed no differences, suggesting similar perceptions between these two cities.

Figure 2 shows the final parameters for each attribute and city, computed as the Porto Alegre baseline plus the corresponding city–attribute interaction. The order of the most penalized and most valued attributes remained relatively stable across cities, reinforcing the importance of factors related to sidewalk conditions, security, and urban attractiveness in shaping pedestrians' experiences. In all locations, these three attributes are among the most sensitive aspects to negative changes, while *Attractiveness* stood out among those showing the greatest improvements. These results suggest that urban interventions aimed at improving some aspects of the environment have a greater potential for positive impact on user perception, whereas shortcomings in these same aspects tend to elicit particularly strong negative reactions.

Table 2. Ordered Logit Models results.

Attribute	coef.	t-value
***Sidewalk conditions* – improved**	0,555	8,870
***Sidewalk conditions* – worsened**	-3,170	-33,339
***Safety* – improved**	-0,037	-0,397
***Safety* – worsened**	-2,078	-22,201
***Security* – improved**	0,394	4,722
***Security* – worsened**	-3,205	-33,873
***Destinations* – improved**	-0,214	-2,351
***Destinations* – worsened**	-1,883	-20,294
***Attractiveness* – improved**	0,202	2,217
***Attractiveness* – worsened**	-2,918	-31,080
DUMMY_BAQ	-0,232	-3,495
DUMMY_CON	-0,290	-3,470
Sidewalk conditions* – improved*: DUMMY_BAQ	-	-
Sidewalk conditions* – worsened*: DUMMY_BAQ	0,623	5,399
Safety* – improved*: DUMMY_BAQ	0,398	3,458
Safety* – worsened*: DUMMY_BAQ	0,647	5,648
Security* – improved: *DUMMY_BAQ	0,412	3,801
Security* – worsened: *DUMMY_BAQ	0,711	6,187
Destinations* – improved: *DUMMY_BAQ	0,535	4,736
Destinations* – worsened: *DUMMY_BAQ	0,435	3,825
Attractiveness* – improved*: DUMMY_BAQ	0,651	5,719
Attractiveness* – worsened: *DUMMY_BAQ	0,430	3,728
Sidewalk conditions* – improved: *DUMMY_CON	-	-
Sidewalk conditions* – worsened: *DUMMY_CON	0,565	3,534
Safety* – improved: *DUMMY_CON	0,739	4,495
Safety* – worsened: *DUMMY_CON	0,839	5,306
Security* – improved: *DUMMY_CON	-	-
Security* – worsened: *DUMMY_CON	0,757	4,637
Destinations* – improved: *DUMMY_CON	1,084	6,782
Destinations* – worsened: *DUMMY_CON	1,774	10,943
Attractiveness* – improved: *DUMMY_CON	1,868	10,856
Attractiveness* – worsened: *DUMMY_CON	1,093	6,833

Log-likelihood:	-35.064,63	**AIC:**	70.205,25	**BIC:**	70.501,78

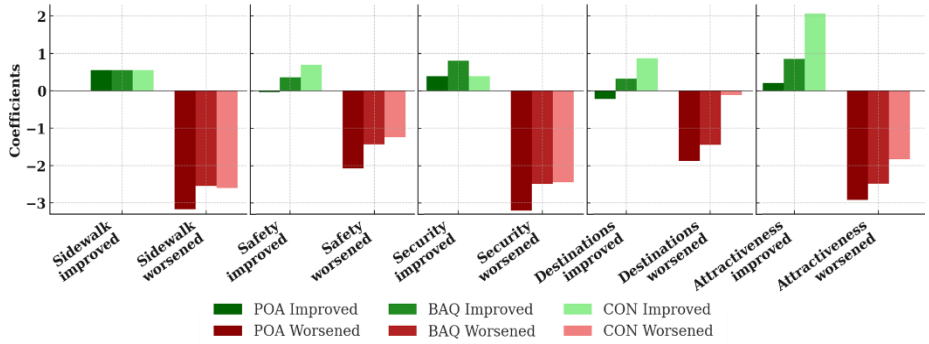


Fig. 2. Attribute coefficients estimated through the Ordered Logit Model

Figure 3 shows the predicted gains and losses in expected scores relative to the baseline scenario, as forecasted by the model for different scenarios of urban improvement and deterioration, derived from the scores assigned by participants in each city. As observed in the statistical analysis, the estimated model results also indicate the existence of an asymmetry in walkability perception, revealing a clear pattern of loss aversion among participants, a phenomenon widely discussed in Prospect Theory in behavioral economics and documented in numerous other domains.

Attributes such as *Sidewalk conditions*, *Security*, and *Attractiveness* showed the steepest declines when deteriorated, while *Attractiveness* and *Destinations* yield the largest gains under improvement, particularly in Concepción. The aggregate comparison ("All improved" vs. "All worsened") clearly highlights this imbalance, with negative changes producing impacts two to three times larger than positive interventions, consistent with the behavioral mechanisms identified earlier.

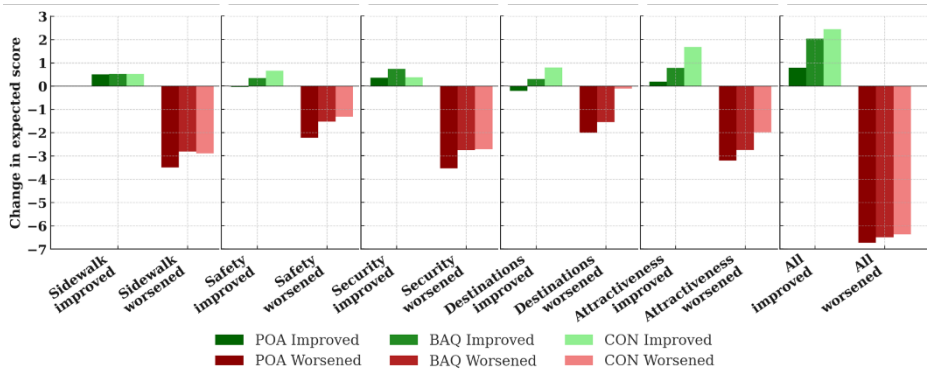


Fig. 3. Impact of improvements and deteriorations in attributes on the scores given to the walkability of the environment

These results also align with previous findings in the literature. Facchini et al. (2025), using a stated preference survey and discrete choice models, identified *Attractiveness* as the attribute with the greatest influence on pedestrian choice, followed by *Sidewalk conditions* and *Security*. While some differences emerge, possibly due to the type

of task, participant profile, or the enhanced realism of the immersive scenarios in the present study, the direction and statistical significance of the effects remain consistent across both investigations. Together, these findings reinforce the pivotal role of these key urban design attributes in shaping pedestrian perception and highlight the importance of prioritizing them in walkability-oriented planning and interventions.

4 Conclusions

This study investigates the presence and magnitude of asymmetries in perceived walkability, exploring whether deteriorations and improvements in key urban attributes exert equivalent effects on pedestrian evaluations. A one-factor-at-a-time (OFAT) experimental design was employed, complemented by descriptive analyses, inferential statistics, and pooled ordered logit modeling. Within the OFAT framework, each attribute was varied independently while the remaining factors were held constant, enabling a controlled assessment of each element's relative contribution to perceived walkability. Furthermore, the use of video-based scenarios enhanced the ecological validity of the experiment, allowing participants to assess urban environments that closely mirrored real-world conditions.

The results indicate significant differences among the three cities analyzed, except for a few specific attributes (*Sidewalks* and *Security* in certain comparisons). Overall, respondents from Barranquilla and Concepción assigned slightly lower scores than those from Porto Alegre, revealing less sensitivity to changes in urban attributes. Sidewalk conditions, security, and urban attractiveness emerged as the most influential factors shaping walkability perceptions, both in terms of gains and losses. Comparing Barranquilla and Concepción, some attributes showed no differences, suggesting similar perceptions between these two cities, while other attributes have different scores (*Safety* (improved), *Destinations* and *Attractiveness* (both improved and worsened)).

In Concepción, improvements in *Attractiveness* and *Destinations* produced the greatest positive impacts, suggesting that in a context with relatively adequate infrastructure and safety, residents value enhancements that make walking more pleasant and purposeful. In Porto Alegre and Barranquilla, the strongest effects came from improvements in *Sidewalks* and *Security,* indicating a greater sensitivity to basic infrastructure and safety conditions. Across all three cities, however, deteriorations in *Security* and *Sidewalks* had the most pronounced negative effects, confirming their central role in shaping walkability perceptions.

A clear asymmetry between improvements and deteriorations was observed: negative changes produced effects two to three times stronger than equivalent positive ones, highlighting a pattern of loss aversion consistent with Prospect Theory. Comparing cities, Porto Alegre tend to exhibit the strongest penalties for deteriorations, while Concepción showed the most significant positive responses.

These findings offer important implications for urban policy and design. They indicate that efforts to enhance walkability should prioritize not only the introduction of positive features but also the prevention of deterioration in key environmental attributes. Furthermore, the study underscores the usefulness of simple and cost-effective

evaluation tools, such as the OFAT-based video experiments employed here, in supporting evidence-based decision-making processes. Future research could expand upon these results by incorporating interactions among multiple attributes, examining behavioral variations across demographic groups, and applying this approach to a broader range of urban and cultural contexts.

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